



Figure 1: PVDM card

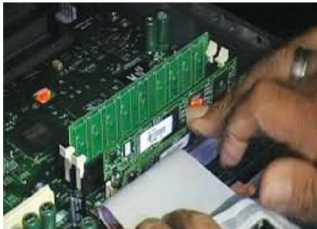


Figure 2: PVDM card installed in Cisco router

M3.bin' with PVDM card v3. PVDM stands for the 'packet voice DSP' (digital signal processing) module. The PVDM card supports voice processing, just like the graphics card supports graphics. Please see Figures 1 and 2 for PVDM cards. Each PVDM v3 card supports 60 calls per second.

c) **E1 card:** In a Cisco router, there is one E1 card which should be installed for the termination of the PRI line to the router.



Note: A Cisco router is used for VXML browsers and MRCP clients.

Server requirements

Two core servers, with a minimum of 2GB of RAM, are sufficient to handle up to 30 calls per second.

Software requirements

- VXML browser:** This stands for Voice XML. It is like form in HTML. It collects data from speech-to-text and posts to the PHP page, or you can also post to the JSP page. It runs in the Cisco IOS browser. You can also install a separate VXML browser in CentOS, which is free and open source, like Voiceglue (a combination of Asterisk and open VXI). But I recommend that you use the Cisco IOS browser.
- MRCP server:** Media Resource Control Protocol (MRCP) is a communication protocol used by speech servers to

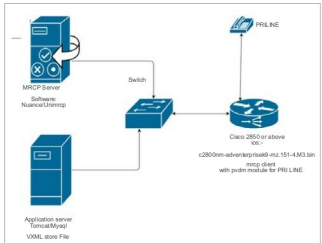


Figure 3: Block diagram for server, router and PRI line

provide various services (such as speech recognition and speech synthesis) to clients. Software for the MRCP server is Unimrcp, or you can use Nuance as the MRCP server. Here, the MRCP client will be the Cisco router. MRCP relies on another protocol, such as the Real Time Streaming Protocol (RTSP) or Session Initiation Protocol (SIP) for establishing a control session and audio streams between the client and the server.

- Database:** A MySQL server is used as the database server.
- Web server:** Apache is the Web server for content delivering systems.
- Server side scripting language:** PHP
- Operating system:** CENTOS 6.7

Please take a look at Figure 3. I will attempt to explain the technologies required to build a speech based IVR system.

Let's first set up the Cisco 2850 router for VXML browsing with a PRI line. Please find the configuration as given below for the Cisco router 2851. I have a PRI line pilot number 01206681400. The pilot number is a telephone number in the range of 6681400 to 6681430. If two or more than two people dial the number 6681400, then the system will connect the first one to 6681400, the second one to 6681401, the third to 6681402, and so on. But the caller will dial only the number 6681400.

To set up a router for receiving calls, the following prerequisites need to be in place:

- IOS version: c2800nm-adventerprisek9-mz.151-4.M3.bin
- E1 card
- PVDM Card v3

Configuring the router

To do this, we have to focus on three steps:

- Setting up the controller
- Creating an application service
- Dial-peer

I am assuming that you know how to set up a router with an IP address and remote access, and are able to define the route.